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$$\begin{aligned}f(x) &= \frac{3}{x-2} + \frac{x}{x-1} \\f'(x) &= \frac{(x-2) \cdot 0 - 3 \cdot 1}{(x-2)^2} + \frac{(x-1) \cdot 1 - x \cdot 1}{(x-1)^2} \\&\Rightarrow f'(x) = \frac{-3}{(x-2)^2} + \frac{x-1-x}{(x-1)^2} \\&\Rightarrow f'(x) = \frac{-3}{(x-2)^2} + \frac{-1}{(x-1)^2} \\&\Rightarrow f'(x) = \frac{-3}{(x-2)^2} - \frac{1}{(x-1)^2}\end{aligned}$$

References